

# Predicting Sleep Quality from Wearable Multisensor Data

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*Group 23, COGS 109 Final Project Report*  
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## Abstract

This project investigates whether daytime wearable sensor data can predict sleep quality using the Multilevel Monitoring of Activity and Sleep in Healthy People (MMASH) dataset. After cleaning the data, we analyzed 21 participants using six predictors: age, BMI, average heart rate, total steps, mean vector magnitude, and heavy activity events. We created a Sleep Quality Score from sleep duration, efficiency, wake after sleep onset, awakenings, and sleep fragmentation. We compared Ordinary Least Squares (OLS) regression, Lasso regression, and a naive mean baseline using 5-fold cross-validation. Although Lasso performed better than OLS, the naive mean baseline had the lowest prediction error overall, suggesting that the selected features were not strong predictors of sleep quality in this small sample. The results indicate that more data and richer sleep and activity features are needed before making reliable predictive or causal conclusions.

## Introduction

Sleep quality is a critical component of overall human health, yet continuous and objective monitoring remains difficult. Traditional clinical methods such as polysomnography are highly accurate, but they are invasive, expensive, and usually restricted to laboratory settings. Wearable technology provides a more accessible alternative for tracking physiological and behavioral metrics in daily life, but converting raw wearable data into an interpretable sleep outcome remains a challenging data science problem. This project addresses that problem by building a model that uses wearable and demographic features to predict a derived continuous Sleep Quality Score.

We use the MMASH dataset, which includes participant sleep records, actigraphy data, heart rate data, beat-to-beat interval data, activity logs, questionnaire scores, and demographic information. The raw dataset contains 22 participant folders. After preprocessing and dropping one participant with incomplete required variables, the final cleaned analysis dataset contains  $n = 21$  observations and  $p = 6$  predictors: age, body mass index (BMI), average heart rate, total daily steps, mean vector magnitude of movement, and number of heavy activity events.

Since the dataset does not provide a single clinical ground-truth sleep quality label, we computed a Sleep Quality Score on a 0 to 100 scale, where higher values indicate better sleep quality. This score is derived from available sleep variables: total sleep time, sleep efficiency, wake after sleep onset (WASO), number of awakenings, and sleep fragmentation index.

## Hypothesis

We hypothesize that participants with higher daytime physical activity will have higher Sleep Quality Scores. We expect `total_steps`, `mean_vector_mag`, and `heavy_activity_events` to be positively associated with the Sleep Quality Score.

Our null hypothesis is that daytime physical activity is not associated with the Sleep Quality Score. This hypothesis is evaluated using both prediction performance and coefficient direction. Prediction performance is measured with cross-validated mean squared error (MSE), and coefficient direction is used only for exploratory interpretation because the sample size is small.

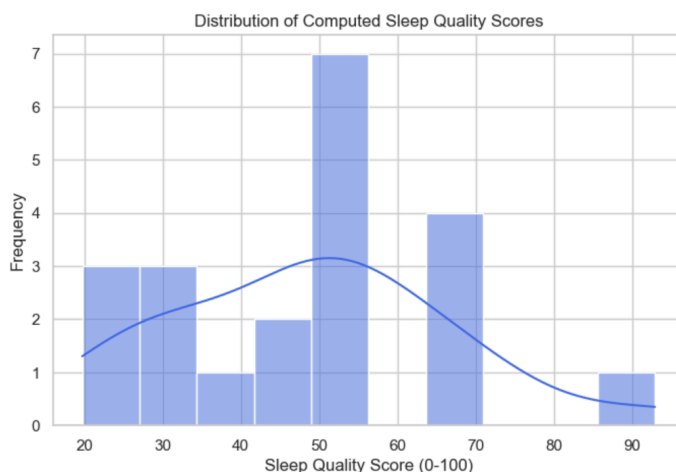
## Methods

### Data Processing:

The cleaned MMASH dataset is structured across 21 distinct participant folders, each containing individual CSV files. The data collection spans a continuous 24-hour window split across two calendar days. Day 1 runs from approximately 09:30 to 23:59, and Day 2 covers 00:00 to approximately 09:30. Our preprocessing method accesses each user folder and chronologically merges the continuous biosignals to accurately capture a single, consolidated wake-sleep cycle per participant.

We aggregate the second-by-second wearable data and categorical activity logs into daytime summary statistics. The dataset does not provide a singular ground-truth sleep score, so we compute a Sleep Quality Score scaled from 0 to 100 based on the recorded metrics. This score represents a weighted linear combination of positive restorative factors (Total Sleep Time, Latency Efficiency) penalized by sleep disruptions (Wake After Sleep Onset, Sleep Fragmentation Index, and Number of Awakenings).

**Figure 1.**



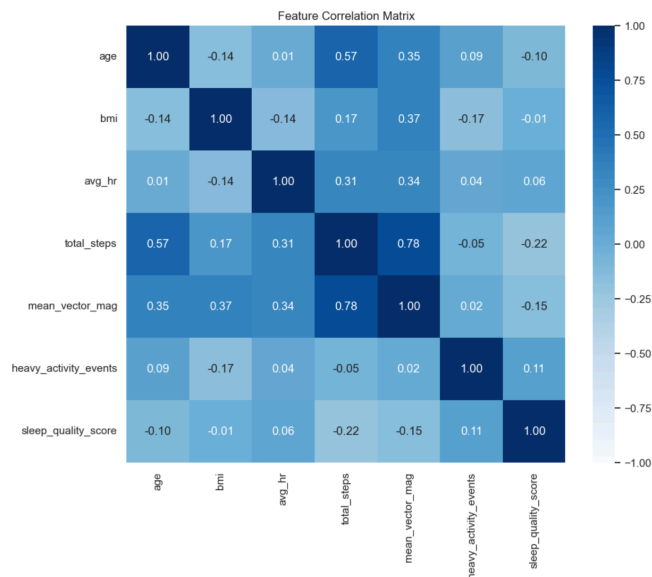
This histogram shows that our calculated Sleep Quality Scores form a roughly normal distribution centered around 50, providing meaningful variation for our regression model.

**Data Analysis Approach & Rationale:**

Our primary analytical approach is Multiple Linear Regression, aiming to predict a continuous sleep quality score based on preceding daytime physiological and behavioral metrics. By examining the resulting beta coefficients, we can infer which specific daytime variables have the strongest statistical association with sleep quality.

We compare results from OLS regression with Lasso regression, as OLS serves as the baseline regression model using all six predictors, while Lasso regression adds an L1 penalty. This is useful since some predictors are correlated. For instance, total\_steps and mean\_vector\_mag have a correlation of approximately 0.78, meaning they both measure related aspects of daytime movement. Lasso is therefore useful for reducing redundant predictors and limiting overfitting in a small dataset. Our linear regression model would be able to define an additive relationship between cumulative daily behaviors and overall sleep restfulness.

**Figure 2.**



There is a strong multicollinearity (0.78) between total steps and mean vector magnitude. This visual evidence justifies our selection of Lasso regression to automatically penalize and shrink these redundant coefficients.

**Model Estimation and Validation Strategy:**

We utilize k-fold cross-validation, specifically 5-fold cross-validation, to ensure our model generalizes well to unseen data and avoids overfitting. We randomly shuffle the order of our aggregated observations across the 21 users to remove any subject-ordering bias, and then split the dataset into five equal-sized subsets. The model estimation process iterates five times. During each of the five iterations, the algorithm trains the regression model on 80 percent of the data and tests its predictive accuracy on the remaining 20 percent of the data. We then calculate

the Mean Squared Error for each subset, and the final reported cross-validation error represents the average of these five individual test errors.

To avoid data leakage, the feature standardization is performed inside the model pipeline during cross-validation. This ensures that the scalar is fit only on the training data within each fold. The Lasso hyperparameter tuning is also handled within the cross-validation process using GridSearchCV over alpha values from 0.001 to 1.0. In addition to OLS and Lasso, we include a naive mean baseline using DummyRegressor. This baseline predicts the training-set mean Sleep Quality Score for each test participant and helps determine whether the regression models actually improve on a simple constant prediction.

## **Results**

### **Model Selection:**

From the analysis ran on  $n = 21$  observations, the calculated Sleep Quality Score ranged from 19.75 to 92.93 (mean = 48.94, SD = 17.52), forming a roughly normal distribution centered near the midpoint of the 0 to 100 scale.

The baseline OLS model produced a mean 5-fold CV MSE of 454.45 (SD = 296.79), corresponding to an RMSE of approximately 21.32 sleep score points. The wide standard deviation across folds (ranging from 146.13 to 854.87) reflects substantial instability, which is consistent with the high predictor-to-observation ratio ( $p/n$ ). Fitting six predictors on only 21 observations leaves OLS highly susceptible to overfitting.

The Lasso model produced a mean 5-fold CV MSE of 361.38 (SD = 275.90), corresponding to an RMSE of approximately 19.01 sleep score points, a 20.5% reduction relative to OLS. Lasso was selected as the preferred model based on this lower CV MSE. The improvement is attributable to Lasso eliminating three of six predictors entirely (age, BMI, and mean\_vector\_mag), removing features that were adding noise rather than signal in a small sample. This is expected when multicollinearity and a high  $p/n$  ratio are present.

Given our small sample size ( $n = 21$ ) relative to the number of predictors ( $p = 6$ ), highly flexible models are prone to overfitting. The cross-validation results reflect this complexity. The flexible OLS model exhibited the highest error (Mean CV MSE: 454.45), indicating it memorized the training folds but generalized poorly. Lasso regression successfully reduced this complexity by shrinking redundant coefficients, improving the Mean CV MSE to 361.38. However, both regression models were outperformed by a naive baseline. The variance of the target variable is approximately 307 ( $SD^2 \approx 17.52^2$ ), and simply predicting the mean score for every participant yields an RMSE of 17.14. The flexible models produced higher RMSEs (19.01 and 21.32), introducing more noise than predictive signal. Neither model can be considered a reliable predictor of sleep quality from these daytime features alone.

### **Model Comparison:**

| Model               | Mean CV MSE | CV MSE Std | RMSE  |
|---------------------|-------------|------------|-------|
| OLS                 | 454.45      | 296.79     | 21.32 |
| Lasso               | 361.38      | 275.90     | 19.01 |
| Naive mean baseline | 293.83      | 211.14     | 17.14 |

Among the two regression models, Lasso performs better than OLS because it has a lower mean CV MSE. However, the naive mean baseline performs best overall. Although Lasso reduces prediction error relative to OLS, neither regression model beats simply predicting the average Sleep Quality Score. This means the six daytime predictors do not provide a strong predictive signal for the calculated sleep score in this small sample. Therefore, the regression models used are exploratory and are not reliable predictors of sleep quality.

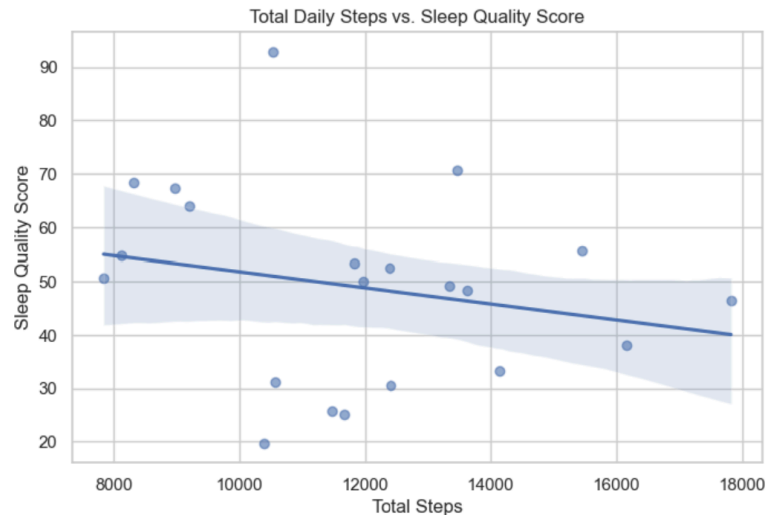
#### **Model Estimation and Coefficient Interpretation:**

| Feature               | Lasso coefficient |
|-----------------------|-------------------|
| age                   | 0.000             |
| bmi                   | 0.000             |
| avg_hr                | 1.077             |
| total_steps           | -3.221            |
| mean_vector_mag       | 0.000             |
| heavy_activity_events | 0.744             |

Lasso regularization retained three predictors with non-zero coefficients. The strongest retained predictor was total\_steps (Beta = -3.22), which showed a negative association with sleep quality. This partially contradicts our hypothesis, as we believed that higher step counts would be positively associated with sleep quality. One possible explanation is that high-activity days may increase physiological activation or accumulate fatigue in ways that disrupt overnight sleep, despite this the small sample size limits any causal interpretation. This negative association was also visually consistent with the scatter plots from our exploratory analysis. Average heart rate (Beta = +1.08) showed a positive association with sleep quality. Rather than reflecting on a heightened stress response, this may indicate that participants with higher baseline cardiovascular activity during the day are generally more physically fit, and fitness can be associated with better sleep outcomes. Heavy activity events showed a small positive association

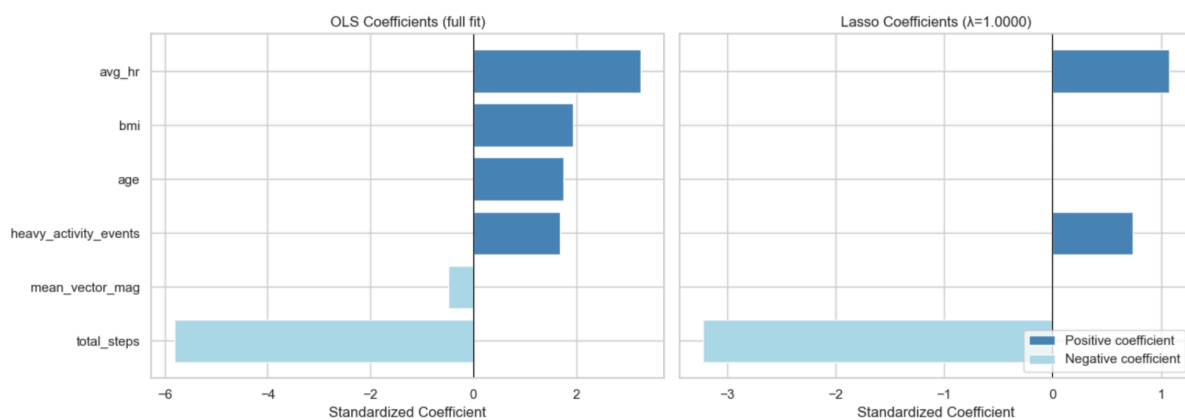
(Beta = +0.74), which partially supports the hypothesis that intense activity is beneficial for sleep quality.

**Figure 3.**



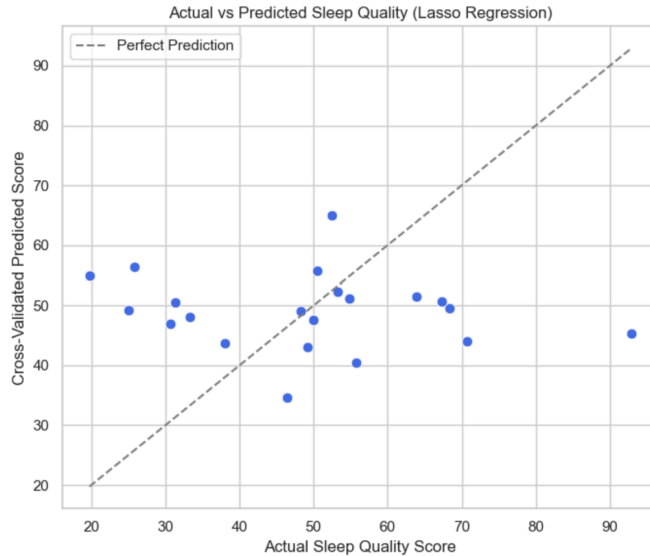
The Total Daily Steps vs. Sleep Quality Score scatter plot depicts the exploratory relationship between total daytime steps and the composite Sleep Quality Score. The slightly negative trendline directly corroborates the negative Lasso coefficient (-3.22) assigned to the step count predictor, reinforcing that higher step counts did not automatically guarantee better sleep outcomes in this sample

**Figure 4.**



The OLS versus Lasso Coefficients bar chart compares the standardized coefficients of the baseline OLS model against our final Lasso model (Alpha = 1.0000). The L1 penalty successfully identified the severe multicollinearity, shrinking the mean\_vector\_mag, age, and bmi coefficients completely to zero.

**Figure 5.**



The Actual vs Predicted scatter plot visualizes the cross-validated predictions of the Lasso model against the actual sleep scores. The relatively flat horizontal cluster of predictions, compared to the widespread of actual scores along the perfect prediction line, visually reinforces why the model struggled to beat the naive mean baseline

## Discussion and Conclusion

Our hypothesis predicted that higher daytime physical activity would be associated with better sleep quality. The results do not provide strong support for this hypothesis. Lasso performed better than OLS, which supports the idea that regularization is helpful when predictors are correlated and the sample size is small. However, the naive mean baseline still produced the lowest cross-validated error, meaning the regression models did not improve on a simple constant prediction.

We can conclude that the derived Sleep Quality Score is not well predicted by these six daytime features alone in this small dataset. The weak prediction may be due to the small number of participants, noise introduced by constructing a composite sleep score, individual differences in sleep physiology, and the possibility that daytime activity features alone are insufficient for predicting sleep quality. The coefficient results should therefore be treated as exploratory rather than confirmatory. The negative coefficient for total\_steps does not prove that walking more causes worse sleep. It only shows that, in this small sample and composite-score framework, higher step count was associated with lower predicted sleep quality after standardization and regularization.

A major limitation of this analysis is the small sample size ( $n = 21$ ). Physiological data is highly noisy, and a sample of this size is insufficient for a machine learning model to separate genuine biological patterns from random variance without severe overfitting. Future work should use a larger participant sample, evaluate the original sleep metrics separately instead of relying only on a composite score, include richer behavioral or physiological features, and use nested cross validation or a separate test set for more reliable model selection. If the goal is inference

rather than prediction, a future analysis should focus on a smaller set of biologically motivated predictors and report confidence intervals and uncertainty more explicitly.

## **References**

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## **GitHub Repository**

[https://github.com/amitine26/COGS109\\_Final\\_Project/tree/main](https://github.com/amitine26/COGS109_Final_Project/tree/main)